

$$P = 208 \text{ kW}$$

7. Ans: A

Work done by the centripetal force is *zero* because there is no displacement in the direction of the centripetal force which is towards the centre of the circle.

In the time of π/ω , the object would have moved half a circle.

Therefore, change in linear momentum, $\Delta p = mv - (-mv) = 2mv = 2mr\omega$